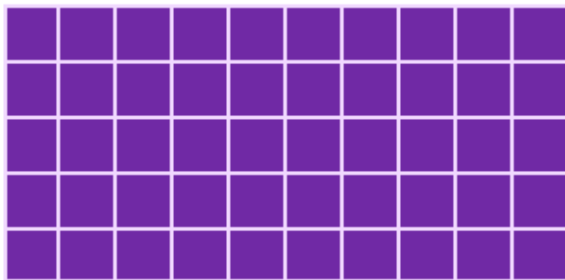
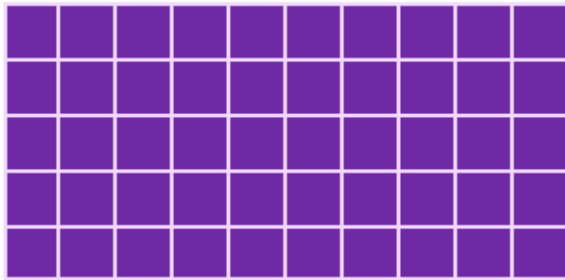


Mini Lesson Sample Script

Materials: Personal white board

1. Find the area of two equal rectangles by labeling side lengths and multiplying.

Label the side lengths of each rectangle. Then write a multiplication equation to find the area of one rectangle. What is the area of the other rectangle? What is the total area of the two rectangles?



Answer: 5 units by 10 units; $5 \times 10 = 50$, 50 square units; 50 square units; 100 square units

T: Take a look at these two rectangles on the grid. Count the squares along each side and label the side lengths.

S: One side is 5 units and the other side is 10 units. Both rectangles have the same side lengths.

T: Write a multiplication equation to find the area of one rectangle.

S: $5 \times 10 = 50$. The area is 50 square units.

T: What's the area of the other rectangle? How do you know?

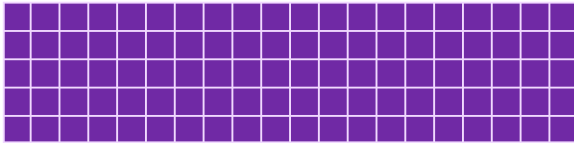
S: It's also 50 square units because the two rectangles are equal.

T: Good. What's the total area of both rectangles together?

S: $50 + 50 = 100$. So 100 square units in all.

2. Find the area of a longer rectangle formed by joining two equal rectangles.

The two equal rectangles from Problem 1 are placed side by side to form one longer rectangle. Label the side lengths of the longer rectangle. Then find its total area.



Answer: 5 units by 20 units; 100 square units

T: Now we push the two equal rectangles together to make one longer rectangle. What are the side lengths of this new rectangle?

S: One side is still 5 units. The long side is $10 + 10$, so it's 20 units.

T: Label them. How could you find the area of the longer rectangle?

S: I could multiply 5×20 . → Or I could add the two areas I already found.

T: Let's try multiplying. What is 5×20 ?

S: 5×20 is 5×2 tens. That's 10 tens, or 100 square units.

T: And if we add the two partial areas?

S: $50 + 50 = 100$ square units. Same answer.

T: Right. The total area of the longer rectangle is the same as the sum of the two smaller rectangles. The square units don't change when we push the rectangles together.

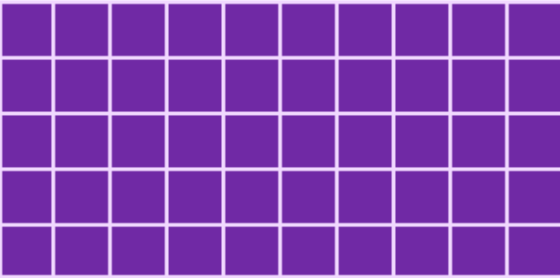
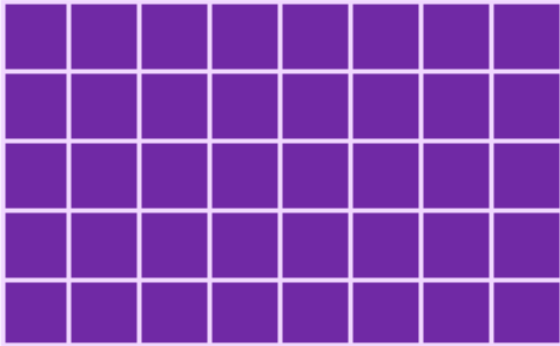
3. Find the area of a longer rectangle formed by joining two unequal rectangles.

Two unequal rectangles are joined along a matching side to form one longer rectangle.

T: Try this one on your own. Find the area of the longer rectangle made from these two.

Label the side lengths of the longer rectangle.

Then find its total area.



Answer: 5 units by 18 units; 90 square units

S: (Work.)

T: How did you find the area, and how did you know the long side was 18 units?

S: I found $5 \times 8 = 40$ and $5 \times 10 = 50$. Then I added $40 + 50 = 90$ square units. The long side is $8 + 10$, so it's 18 units.